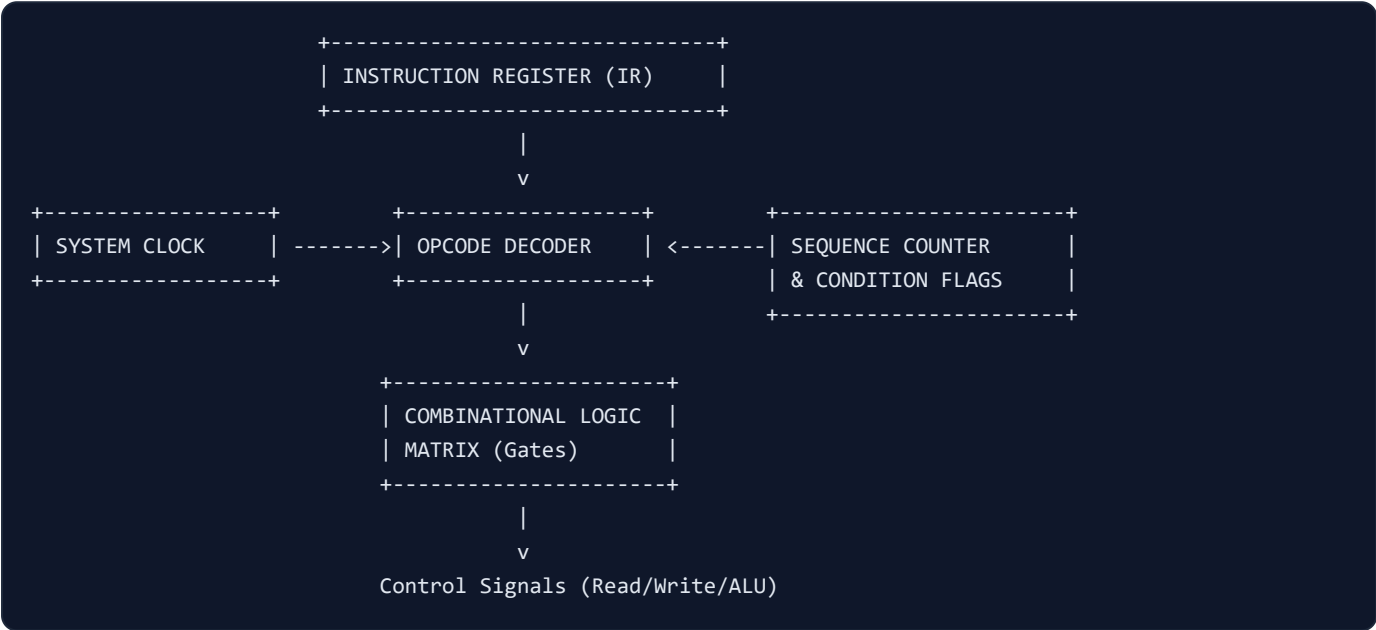


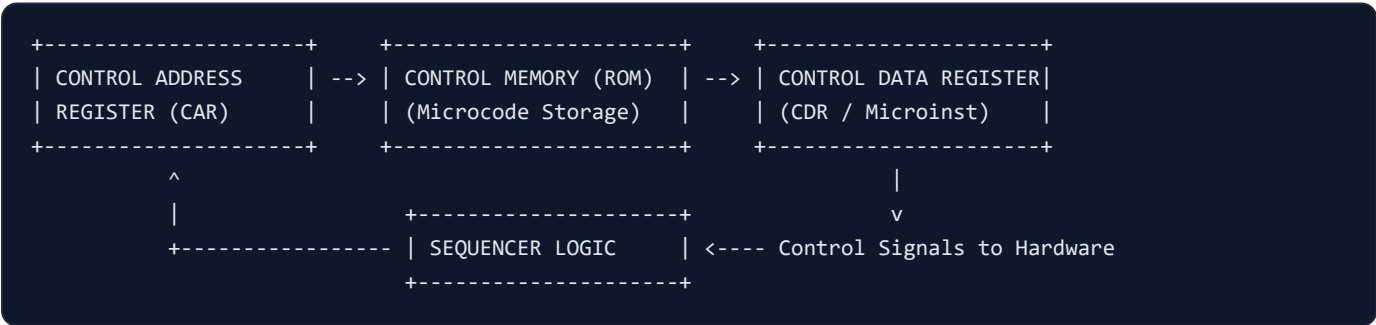
10-Mark Questions & Answers — MPCA Unit 2: Processor Organization & Architecture

Q1. Explain hardwired and microprogrammed control units in detail with block diagrams, and compare their advantages/disadvantages.

1. Hardwired Control Unit Architecture



2. Microprogrammed Control Unit Architecture



3. Comprehensive Comparison Table

Feature	Hardwired Control Unit	Microprogrammed Control Unit
Logic Implementation	Fixed combinational logic gates & flip-flops.	Microcode software stored in Control Memory ROM.
Execution Speed	Extremely fast (limited only by hardware gate delay).	Slower (dependent on Control ROM fetch cycle time).
Flexibility	Rigid. Modifications require hardware redesign.	Highly flexible. Upgraded by updating Control ROM microcode.
Instruction Set Type	Best suited for RISC architectures.	Best suited for complex CISC architectures.
Cost & Complexity	High cost and design complexity for large ISAs.	Systematic design, lower cost for complex ISAs.
Debugging & Maintenance	Difficult to debug and maintain.	Easier to debug and maintain.

Q2. Explain instruction pipelining in detail. Discuss structural, data, and control hazards with examples and methods to resolve each (forwarding, stalling, branch prediction).

1. 5-Stage RISC Instruction Pipeline

```

Stage 1: IF (Instruction Fetch)  -> Read instruction from I-Cache.
Stage 2: ID (Instruction Decode) -> Read registers & decode instruction.
Stage 3: EX (Execution / ALU)   -> Execute ALU operation or calculate memory address.
Stage 4: MEM (Memory Access)   -> Read/Write data from D-Cache.
Stage 5: WB (Write Back)       -> Write result back into destination register.

```

2. Pipeline Hazards & Resolution Strategies

1. Structural Hazards (Resource Contention):

- *Example:* A single-port memory attempting simultaneous instruction fetch (**IF**) and data memory read (**MEM**).
- *Resolution:* Implement separate Instruction and Data Caches (Harvard Architecture model).

2. Data Hazards (Data Dependencies):

- *Example (RAW Hazard):*

```
assembly ADD R1, R2, R3 ; R1 written in WB stage SUB R4, R1, R5 ; R1 read in ID stage (Hazard!)
```

- *Resolution Methods:*

- **Data Forwarding (Bypassing):** Connect the ALU output of **EX** stage directly back to **EX** stage input via a multiplexer, eliminating wait cycles.
- **Pipeline Stalling (NOPs / Bubbles):** Insert hardware stall cycles until data is written back.

3. Control Hazards (Branch Dependencies):

- *Example:* Conditional branch instruction **BEQ R1, R2, Target** alters sequential instruction fetching.
- *Resolution Methods:*

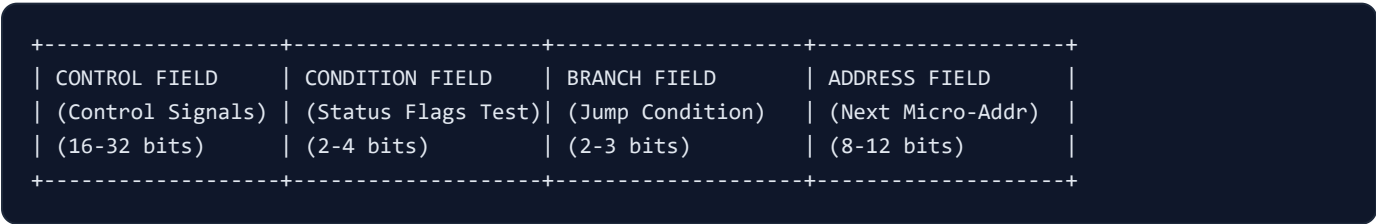
- **Static/Dynamic Branch Prediction:** Predict whether branch is taken or not taken (Branch History Table).
- **Branch Target Buffer (BTB):** Cache branch target addresses for instant fetching.

Q3. Explain the basic instruction cycle with interrupt processing in detail, including the fetch-decode-execute cycle and interrupt handling mechanism.



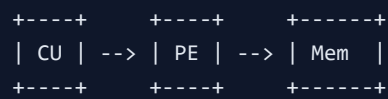
Q4. Describe microinstruction sequencing and execution techniques (in microprogrammed control) along with a suitable example of microinstruction format.

1. Microinstruction Format Structure

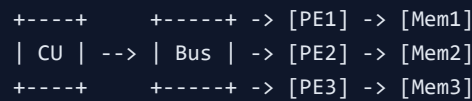


Q5. Explain Flynn's classification of computer architectures in detail with block diagrams and real-world examples for each category.

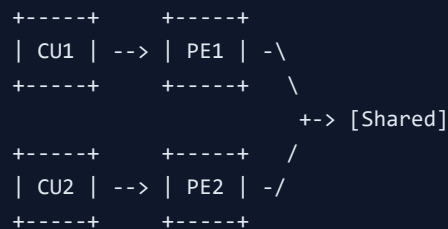
1. SISD Architecture



2. SIMD Architecture



3. MISD Architecture



4. MIMD Architecture

